

## **EXPERIMENT 15**

### **Decision Tree**

#### **Aim**

To implement a **Decision Tree classification algorithm** using Python.

#### **Algorithm**

1. Store the training data.
2. Calculate the best feature for splitting the data.
3. Split the data into branches.
4. Repeat the process until a decision can be made.
5. Give a new input to the decision tree.
6. Predict the output class.
7. Display the result.

#### **Program**

```
# Decision Tree without sklearn
```

```
# Training data
```

```
# [Study Hours, Attendance]
```

```
data = [
```

```
    [2, 60, "Fail"],
```

```
    [3, 65, "Fail"],
```

```
    [4, 70, "Pass"],
```

```
    [5, 75, "Pass"],
```

```
    [6, 80, "Pass"],
```

```
[7, 85, "Pass"]  
  
]  
  
# Function to predict result  
  
def decision_tree(study_hours, attendance):  
  
    if study_hours <= 3:  
  
        return "Fail"  
  
    elif attendance < 70:  
  
        return "Fail"  
  
    else:  
  
        return "Pass"  
  
  
# Input  
  
study_hours = int(input("Enter study hours: "))  
attendance = int(input("Enter attendance percentage: "))  
  
# Prediction  
  
result = decision_tree(study_hours, attendance)  
  
# Output
```

```
print("\nStudy Hours:", study_hours)
```

```
print("Attendance:", attendance)
```

```
print("Predicted Result:", result)
```

## Input

Enter study hours: 5

Enter attendance percentage: 75

## Output

```
Enter study hours: 5
Enter attendance percentage: 75

Study Hours: 5
Attendance: 75
Predicted Result: Pass
```

## Result

Thus, the **Decision Tree classification algorithm** was successfully implemented in Python without using external libraries.

